AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

Physics Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

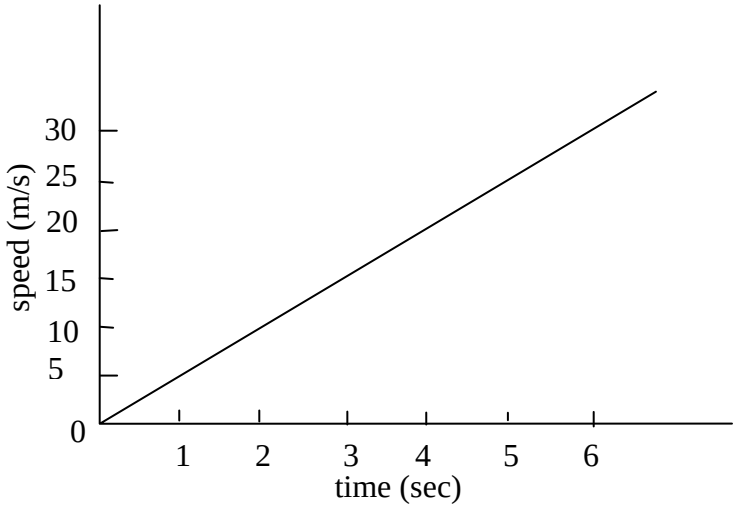
Q.1. (Total 3 Marks)

Write THREE differences between Base and Derived physical quantities.

S. No.	Base quantities	Derived quantities
1.		
2.		
3.		

Q.2. (Total 6 Marks)

Given below is the speed-time graph for a motorcycle travelling along a straight road.



Time in sec	0	1	2	3	4	5	6
Speed in m/s	0	5	10	15	20	25	30

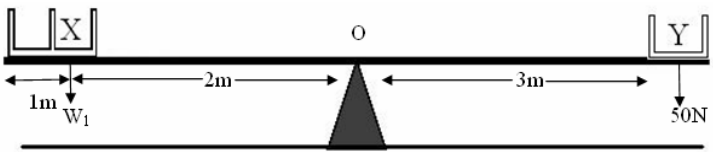
a. What is the distance travelled by the motorcycle in 6 seconds? (2 Marks)

b. What is the acceleration of the motorcycle? (2 Marks)

c. What does the gradient of speed-time graph show? (1 Mark)

d. What does the area under the line of speed-time graph show? (1 Mark)

Q.3. (Total 3 Marks)



The above figure shows a balanced see-saw where a student is seated at position X, while his friend is seated at position Y.

a. State the principle of moment. (1 Mark)

_____

_____

b. What will be the weight of the student? (2 Marks)

_____

_____

_____

_____

Q.4. (Total 3 Marks)

If the Earth pulls an object of mass 100 gm with a force of 1 N, then the same object on the Moon will have less weight of about 1/6 N.

a. What will be the weight of that object on the Moon? (2 Marks)

_____

_____

_____

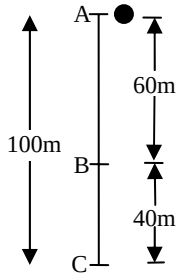
_____

b. Why is the weight of that object on the Moon less than on the Earth? (1 Mark)

_____

_____

Q.5. (Total 6 Marks)



A ball of mass 90 gm dropped from point A passes through point B. Calculate the total energy of the ball at point B.

Q.6. (Total 4 Marks)

A vacuum flask is made of two glass walls with a near-vacuum in between. The bottle is used to maintain liquids, and sometimes solids, at near-constant temperatures. An efficient flask can keep its contents at the desired hot or cold temperature for as long as three days. This is achieved by preventing the transfer of heat between the bottle and its surroundings. The walls are usually made of glass and painted with a reflective metal to reduce the transfer of heat by radiation. The vacuum in the walls greatly reduces the transfer of heat by convection.

a. Write any ONE way to stop heat transfer by convection in the vacuum flask. (1 Mark)

_____

_____

b. Why are the walls of the flask painted with reflecting materials? (1 Mark)

_____

_____

c. Identify any ONE negative effect, if the walls of the flask are made of copper. (1 Mark)

_____

_____

d. How is radiation different from conduction and convection? (1 Mark)

_____

_____

Q.7. (Total 5 Marks)

EITHER

a. Derive Newton's third equation of motion for a body moving with uniform acceleration **'a'** on a straight path.

OR

b. Name and describe the external forces acting on the following objects:

(i) An object is placed on an inclined plane.
(ii) A satellite is revolving around the Earth.

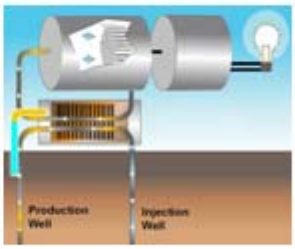
Q.8. (Total 5 Marks)

EITHER

a. A lift of mass 600 kg carries a load of 90 kg at a height of 15 m in 10 sec. Calculate the work done in lifting the load and the power of the motor driving the lift.

OR





b. Recognize the above renewable energy sources and describe any TWO of them.

Q.9. (Total 5 Marks)

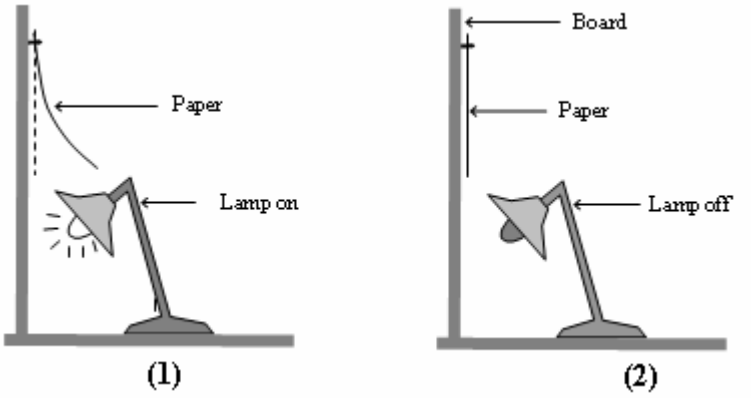
EITHER

a. Liquid left in air with no rise in temperature slowly changes into vapours. This process is known as evaporation.

Identify and describe any TWO factors affecting the rate of evaporation.

OR

b. In the given diagram, an electric lamp is placed directly above a paper board, on which a piece of paper is pinned along its top edge.



Explain what happens when the lamp is switched on and off.

END OF PAPER

Please use this page for rough work

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